

Name _____

Period _____ Date _____

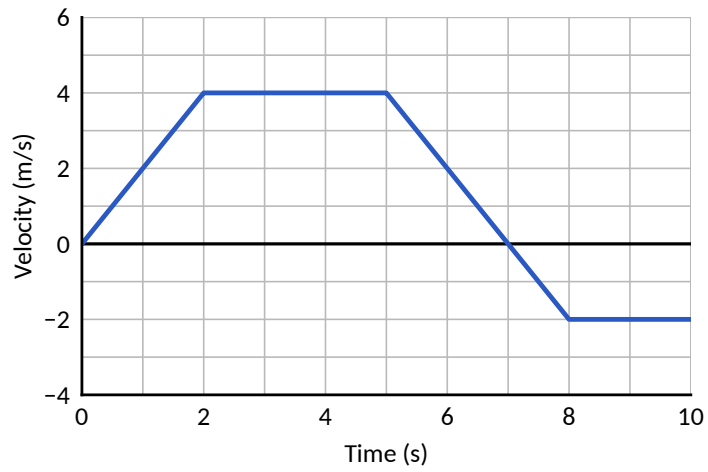
Practice Test A – Advanced Kinematics

On any problems, you must show all work, include units, round properly, and box your final answer.

Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

(25%) The velocity-time graph below shows the motion of a cart on a straight track over the course of 10 seconds. Sketch the corresponding acceleration-time graph.



(25%)

- How long is the marble in the air?
- How far from the base of the table does the marble hit the floor?
- What are the magnitude and direction of the marble's velocity right before it hits the floor?

A student standing on the flat roof of a building 20.0 m above the ground throws a ball straight up at 12.0 m/s. On the way down, the ball just misses the roof and falls to the ground. Define up as the positive direction.

(25%)

- How high above the roof does the ball rise?
- What is the ball's velocity right before it hits the ground?
- How long is the ball in the air?

(25%)

- a. Derive an expression for the projectile's range R in terms of v_0 , θ_0 , and physical constants, as appropriate.
- b. Using your expression from part (a), determine the launch angle that gives the maximum range. Justify your answer.

Answer Key – Practice Test A

Fully worked. Algebra first, numbers last, units inside every parenthesis.

1. Sketch the a-t graph (25%)

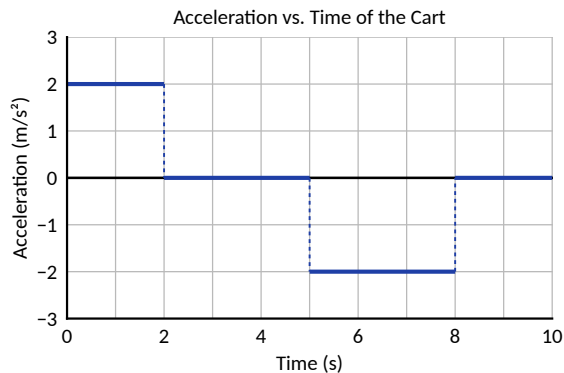
Acceleration = slope of the v-t graph. Find the slope of each straight piece:

$$0 \text{ to } 2 \text{ s: } a = \frac{\Delta v}{\Delta t} = \frac{(4 \text{ m/s} - 0 \text{ m/s})}{(2 \text{ s} - 0 \text{ s})} = +2.0 \text{ m/s}^2$$

$$2 \text{ to } 5 \text{ s: flat line, } a = 0 \text{ m/s}^2$$

$$5 \text{ to } 8 \text{ s: } a = \frac{(-2 \text{ m/s} - 4 \text{ m/s})}{(8 \text{ s} - 5 \text{ s})} = \frac{(-6 \text{ m/s})}{(3 \text{ s})} = -2.0 \text{ m/s}^2$$

$$8 \text{ to } 10 \text{ s: flat line, } a = 0 \text{ m/s}^2$$



Watch out: the cart turns around at 7 s ($v = 0$), but the acceleration is still -2.0 m/s^2 there. $v = 0$ does not mean $a = 0$.

2. Marble off the table (25%)

horizontal	vertical
$a_x = 0$	$a_y = -g$
$v_{x0} = v_0 = 2.40 \text{ m/s}$	$v_{y0} = 0$
$\Delta x = ?$	$\Delta y = -h = -1.25 \text{ m}$

$$(a) \Delta y = v_{y0}t + \frac{1}{2}a_y t^2$$

$$-h = 0 - \frac{1}{2}gt^2 \rightarrow t = \sqrt{\frac{2h}{g}}$$

$$t = \sqrt{\frac{2(1.25 \text{ m})}{(9.8 \text{ m/s}^2)}} = \boxed{0.505 \text{ s}}$$

$$(b) \Delta x = v_{x0}t + \frac{1}{2}a_x t^2 = v_0 t = v_0 \sqrt{\frac{2h}{g}}$$

$$\Delta x = (2.40 \text{ m/s})\sqrt{\frac{2(1.25 \text{ m})}{(9.8 \text{ m/s}^2)}} = \boxed{1.21 \text{ m}}$$

$$(c) v_x = v_0 = 2.40 \text{ m/s (never changes, } a_x = 0)$$

$$v_y = v_{y0} + a_y t = -gt = -g\sqrt{\frac{2h}{g}} = -\sqrt{2gh} = -\sqrt{2(9.8 \text{ m/s}^2)(1.25 \text{ m})} = -4.95 \text{ m/s}$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(2.40 \text{ m/s})^2 + (-4.95 \text{ m/s})^2} = \boxed{5.50 \text{ m/s}}$$

$$\theta = \tan^{-1} \frac{|v_y|}{v_x} = \tan^{-1} \frac{(4.95 \text{ m/s})}{(2.40 \text{ m/s})} = \boxed{64.1^\circ \text{ below the horizontal}}$$

3. Ball thrown up from the roof (25%)

Up is +, origin at the roof. $a = -g = -9.8 \text{ m/s}^2$

(a) At the top $v = 0$. No time given or asked, so use $v^2 = v_0^2 + 2a\Delta y$

$$0 = v_0^2 - 2g\Delta y \rightarrow \Delta y = \frac{v_0^2}{2g} = \frac{(12.0 \text{ m/s})^2}{2(9.8 \text{ m/s}^2)} = \boxed{7.35 \text{ m}}$$

(b) Ground is 20.0 m below the roof, so $\Delta y = -h = -20.0 \text{ m}$

$$v^2 = v_0^2 + 2(-g)(-h) = v_0^2 + 2gh \rightarrow v = -\sqrt{v_0^2 + 2gh} \text{ (negative root: moving down)}$$

$$v = -\sqrt{(12.0 \text{ m/s})^2 + 2(9.8 \text{ m/s}^2)(20.0 \text{ m})} = \boxed{-23.2 \text{ m/s}} \text{ (23.2 m/s downward)}$$

(c) Now that v is known, use $v = v_0 + at$, no quadratic needed:

$$t = \frac{v - v_0}{-g} = \frac{v_0 + \sqrt{v_0^2 + 2gh}}{g} = \frac{(12.0 \text{ m/s}) + (23.2 \text{ m/s})}{(9.8 \text{ m/s}^2)} = \boxed{3.59 \text{ s}}$$

Watch out: the landing speed is not 12.0 m/s. It only matches the launch speed when the ball lands at the height it was thrown from.

4. Range derivation (25%), variables only

horizontal	vertical
$a_x = 0$	$a_y = -g$
$v_{x0} = v_0 \cos\theta_0$	$v_{y0} = v_0 \sin\theta_0$
$\Delta x = R$	$\Delta y = 0$ (lands at launch height)

(a) Time from the vertical: $\Delta y = v_{y0}t + \frac{1}{2}a_y t^2$

$$0 = (v_0 \sin\theta_0)t - \frac{1}{2}gt^2 = t(v_0 \sin\theta_0 - \frac{1}{2}gt)$$

$$t = 0 \text{ (launch)} \text{ or } t = \frac{2v_0 \sin\theta_0}{g}$$

Horizontal: $R = v_{x0}t + \frac{1}{2}a_x t^2 = (v_0 \cos\theta_0)t$. Substitute the expression for t , not a number:

$$R = (v_0 \cos\theta_0) \frac{2v_0 \sin\theta_0}{g} = \frac{v_0^2 (2 \sin\theta_0 \cos\theta_0)}{g}$$

$$2 \sin\theta \cos\theta = \sin 2\theta, \text{ so } \boxed{R = \frac{v_0^2 \sin 2\theta_0}{g}}$$

(b) v_0 and g are fixed, so R is largest when $\sin 2\theta_0$ is largest. The maximum value of sine is 1, at 90° .

$$2\theta_0 = 90^\circ \rightarrow \boxed{\theta_0 = 45^\circ}$$

Units check: $(\text{m/s})^2 \div (\text{m/s}^2) = \text{m}$.